

Straightening Laws in Algebraic Complexity

Anakin Dey

Section 1

Background

What is Algebraic Complexity?

Question

How hard is it to compute polynomials?

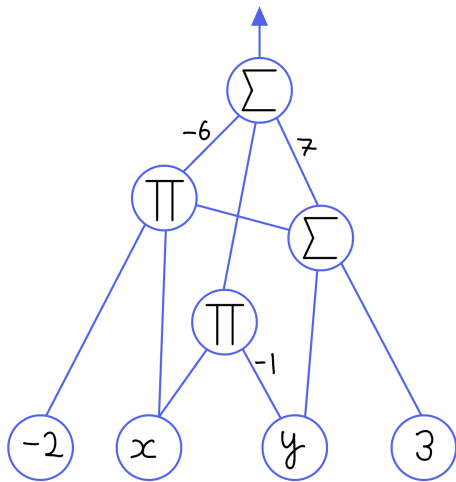
Question

How hard is it to compute polynomials in restricted models?

Question

Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

Horner's Rule

Hard Polynomials Exist

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$, we take a parameter ε , allow our coefficients to be rational functions in ε , and let ε tend to 0.

Definition

We say a circuit *border computes* $g(\bar{x})$ over $\mathbb{F}$ if it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}(\varepsilon)[\bar{x}], \quad q \geq 1.$$

We **cannot** set $\varepsilon = 0$ since we may divide by ε .

What does Border Computation Get Us?

Question

What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the *circuit complexity* of some family of polynomials.

Definition

Fix some polynomials $g_1(\bar{x}), \dots, g_k(\bar{x}) \in \mathbb{F}[\bar{x}]$.

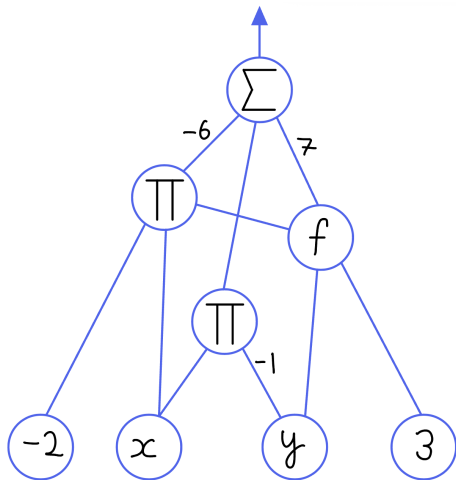
The *ideal* generated by $g_1(\bar{x}), \dots, g_k(\bar{x})$ is the set of combinations

$$\langle g_1, \dots, g_k \rangle = \left\{ \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}) \mid h_i(\bar{x}) \in \mathbb{F}[\bar{x}] \right\}.$$

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Algebraic Circuits with Oracles



Question

How can we compare the *relative* complexity of two polynomials?

We can allow a fixed polynomial $f(\bar{x})$ to be computed with unit cost by allowing it to function as its own gate.

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial g .

If $f \in \langle g \rangle$, then g is a *factor* of f .

Question

Suppose $f \in \langle g \rangle$. Does g have a small f -oracle circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If g is a factor of f , $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

Question

If $f(\bar{x})$ has a small circuit, do $g_1(\bar{x})$ or $g_2(\bar{x})$ have a small f -oracle circuit?

This is open for *general* polynomials $g_1(\bar{x}), g_2(\bar{x})$ [Gro20].

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Prior Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = n^{\Theta(1)}$.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = n^{\Theta(1)}$.

- There exist polynomial size circuits for computing $\det(X)$.
- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Homogenization* as well as a new application of the *Isolation Lemma*.

Section 2

Preliminaries

Bideterminants

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$. Consider the *bitableau* $(S \mid T)$ of shape σ

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right).$$

Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} (x_{4,3}).$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

$$\prod_{i=1}^d x_{r_i, c_i} = \left(\begin{array}{c|c} \boxed{r_1} & \boxed{c_1} \\ \boxed{r_2} & \boxed{c_2} \\ \boxed{\vdots} & \boxed{\vdots} \\ \boxed{r_d} & \boxed{c_d} \end{array} \right) (X).$$

Thus, the bideterminants span $\mathbb{F}[X]$.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X),$$

where A and B are *conjugate semistandard*.

Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

Proposition ([BC03, Corollary 1.7])

The ideal $I_{n,m,r}^{\det}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r .

Shuffle Products

Laplace Expansion

Example Straightening

Proof of Grochow's Conjecture Given Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Theorem

Theorem

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(n^2 m^2 d^3 (n^3 + m^3))$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Let $\sigma = (4, 2, 1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \right) (X)$$

= product of upper-left minors.

Section 3

Proof Techniques

Proof Outline

Since we are starting with $f(X) \in I_{n,m,r}^{\det} \subseteq \mathbb{F}[X]$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X).$$

Our goal will be to use $f(X)$ as an oracle to compute a canonical bideterminant.

To do this, we will isolate one of the terms $(S_k \mid T_k)(X)$.

Linear Operators

Definition (Substitution operator)

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

$$\text{Sub}_{2 \rightarrow 4} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & & \\ \hline 1 & & & \\ \hline \end{array}$$

Let $h_i^j(S)$ be the number of times i is replaced by j in S .

Linear Operators

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\left(\begin{array}{c} \text{Sub} \\ n-1 \rightarrow n \end{array} \circ \begin{array}{c} \text{Sub} \\ n-2 \rightarrow n \end{array} \circ \begin{array}{c} \text{Sub} \\ n-2 \rightarrow n-1 \end{array} \circ \cdots \right. \\ \left. \circ \begin{array}{c} \text{Sub} \\ 2 \rightarrow n \end{array} \circ \cdots \circ \begin{array}{c} \text{Sub} \\ 2 \rightarrow 3 \end{array} \circ \begin{array}{c} \text{Sub} \\ 1 \rightarrow n \end{array} \circ \cdots \circ \begin{array}{c} \text{Sub} \\ 1 \rightarrow 2 \end{array} \right)(S) = \overline{K}_\sigma.$$

Linear Operators

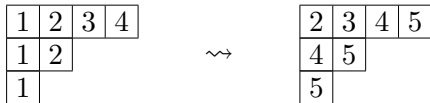
Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\begin{aligned} & \left(\underset{n-1 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n-1}{\text{Sub}} \circ \cdots \right. \\ & \quad \left. \circ \underset{2 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{2 \rightarrow 3}{\text{Sub}} \circ \underset{1 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{1 \rightarrow 2}{\text{Sub}} \right)(S) = \overline{K}_\sigma. \end{aligned}$$

Example

Say $n = 5$



Applying the Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}$, adding a multiple of row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau. Then we have that

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i \rightarrow j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying the Linear Operators

We have seen now that after applying a sequence of linear transformations $E_{i,j}$ to our matrix of variables X , we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

Lemma

$$(\overline{K}_\sigma \mid \overline{K}_\sigma)(J_n X J_n) = \pm(K_\sigma \mid K_\sigma)(X).$$

We now have linear transformations sending bitableau to canonical semistandard Young tableau.

Where Does Isolation Arise?

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X) \in I_{n,m,r}^{\det}.$$

Corollary

We have matrices M, N such that

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X),$$

Upshot: linear transformations correspond to plugging in linear equations into the input for $f(X)$, so this is the start of our f -oracle circuit.

Where Does Isolation Arise?

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X) \in I_{n,m,r}^{\det}, \quad \deg(f) = d.$$

Corollary

We have matrices M, N such that

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X),$$

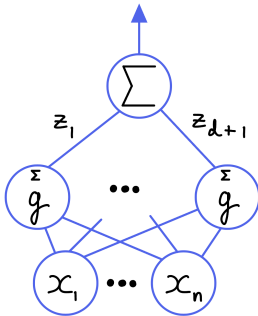
Homogenization

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

Proof of Homogenization: Interpolation

Let $\alpha_1, \dots, \alpha_{d+1}$ be $d + 1$ distinct elements in $\mathbb{F}$.

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \text{coeff}_{t^d}(f) \\ \text{coeff}_{t^{d-1}}(f) \\ \vdots \\ \text{coeff}_t(f) \\ \text{coeff}_1(f) \end{pmatrix} = \begin{pmatrix} f(\bar{x}, \alpha_1) \\ f(\bar{x}, \alpha_2) \\ \vdots \\ f(\bar{x}, \alpha_d) \\ f(\bar{x}, \alpha_{d+1}) \end{pmatrix}$$

This left matrix is a *Vandermonde* matrix and is invertible since the α_i are all distinct.

Proof of Homogenization: Interpolation

$$\begin{pmatrix} \text{coeff}_{t^d}(f) \\ \text{coeff}_{t^{d-1}}(f) \\ \vdots \\ \text{coeff}_t(f) \\ \text{coeff}_1(f) \end{pmatrix} = \begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,d} & z_{1,d+1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,d} & z_{2,d+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{d,1} & z_{d,2} & \cdots & z_{d,d} & z_{d,d+1} \\ z_{d+1,1} & z_{d+1,2} & \cdots & z_{d+1,d} & z_{d+1,d+1} \end{pmatrix} \begin{pmatrix} f(\bar{x}, \alpha_1) \\ f(\bar{x}, \alpha_2) \\ \vdots \\ f(\bar{x}, \alpha_d) \\ f(\bar{x}, \alpha_{d+1}) \end{pmatrix}$$

$$\text{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\bar{x}, \alpha_j).$$

Issues with Homogenization

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon), \quad q \geq 1.$$

Idea: Homogenize with respect to ε .

Problem: q can be *arbitrarily large*

$\implies$ Homogenization gives *large circuit*.

Idea: Kronecker Substitution

Say there is a specific coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$ we want to *isolate*.

Lemma

Suppose that $\deg(g) \leq d$, then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

Problem: The resulting polynomial has *large degree*.

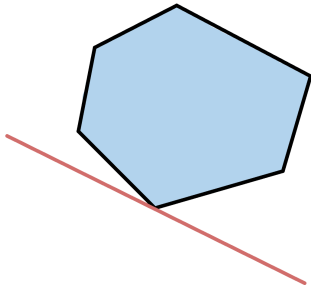
$\implies$ Homogenization gives *large circuit*.

Isolation Lemma

Say there is a specific coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e}} c_{\bar{e}} \bar{x}^{\bar{e}}$ we want to *isolate*.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \geq K\ell/\varepsilon$.

Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \dots, z_\ell$.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be choose to be small
 $\implies$ unique lowest $\deg_w$ -term and homogenization w.r.t w is small.

Section 4

Related and Future Work

Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix, then $\det(X)$ is the perfect square of another polynomial called the *Pfaffian* of X .

Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n, 2r}^{\text{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ pfaffian for some $t = \Theta(r^{1/3})$.

Symmetric Analogue?

The focus on determinantal ideals and pfaffian ideals stem from *invariant theory*. Are there other natural settings to study from there?

Question

Is there an analogue to our results for the ideal of determinants of a symmetric matrix?

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

Question

Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

Thank You!

Algebra is generous; she often gives more than is asked of her.

— Jean Le Rond d'Alembert

Slides can be found on my site anakin.phd

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